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Metadata has long been integral to library cataloging and remains ever important to this day. As time passed and technologies emerged, librarians began to use legacy metadata schemas in use cases that it was not originally intended for. These schemas, such as Dublin Core and MARC, have the capability to be flexible for uses other than literary works, but in many cases fall short. One such example of this is metadata for film or moving images, which could benefit from a universal metadata schema. This report takes a close look at film metadata created with controlled vocabulary in a library system and compares it to metadata for the same film in a specialty database created for films that also allows user input and descriptive tags.

Starting with the Burbank Public Library (BPL), an open catalog which utilizes controlled vocabulary and is run by information professionals, I searched their catalog and narrowed the results using the filters "visual materials", "English language" and "feature films". BPL's catalog does not have great search functions, and this filtering is necessary if a patron is exclusively looking to check out a physical copy of a movie from the library. Using the search bar for a film title without filtering brings mixed results that require additional scrolling and filtering regardless; the BPL does not have the best searchability for this type of item within its online catalog. I chose to search BPL to start because I knew they would have a relatively limited number of films to choose from. The BPL Catalog showed 476 results from this search, which presumes that is the number of physical copies of English language feature films they have in their catalog. After looking at every title within the catalog, eight films were chosen to be included in this metadata comparison. These films, along with their metadata, are listed in Exhibit A.

The criteria used to select the films was popular or cult films that spanned multiple decades. The reason for these criteria was the assessment that popular or cult films have had much written about them and that there would be rich metadata to use for comparison. I also attempted to find film titles with a diverse aspect to them, and while my filter of requiring the English language may have prevented some titles from appearing in my search, the 476 results were limited in their diverse selection.

The second database used for comparison was Internet Movie Database, or IMDB.com. IMDB includes nearly any and every film imaginable in its database. It is essentially a database made exclusively to host the metadata of films. This database is owned by Amazon and carries a staff to keep it updated; these employees are not information professionals and do not necessarily use a controlled vocabulary (IMDB). Due to their ownership, they are partnered with many studios to make sure certain credits are correct. This includes credits such as writing credits in partnership with the WGA (Writers Guild of America) and directing credits with the DGA (Directors Guild of America). Users are also able to submit their own credits or additional information about film titles on the site. These submissions are reviewed by staff for approval before they are published. In addition, the site allows users to add tags and reviews for any film on their site which does not have a formal approval process. It is a joint effort between staff and users to keep the database as accurate and up to date as possible.

With the list from BPL, I searched for the film titles I had chosen on IMDB. The first thing to note about IMDB is that has a lot of ads. It is somewhat hard to navigate on a page if you do not already know what you are looking for; scrolling deep into the page is required to find all the metadata available for each film. However, IMDB has many other advantages. Searching on BPL catalog is a bit clunky and IMDB is more straight forward. The search bar on IMDB has predictive searches, which allows the user to potentially find their chosen film from the search bar without having to click search and look through the search results. Furthermore, IMDB links between every entity. The page for the film The Princess Bride lists all cast, crew, producers, etc, and each name or company is a hyperlink that links to that entities page. Visiting the page for The Princess Bride can link you to the filmography of Robin Wright, which can then link you to the film Forrest Gump, which can link you to the actor Tom Hanks, and so on. Every page is linked to hundreds of other pages which makes the metadata available and access points to any user endless. The BPL catalog entries are siloed and in no way connected to each other.

As previously stated, IMDB does not use a controlled vocabulary the way that BPL does for its metadata, but it does utilize a standard schema that is easy to understand and use for its data entries. BPL metadata is more standardized, but lacks necessary fields needed for film metadata, and therefore not as useful. Fig. 1 shows an example from the BPL website. Much of the important metadata on BPL's search is presented into the searchable "title", which makes the titles very long and overwhelming to read. Once you click into an entry within the BPL catalog, the fields for the metadata do not match the type of data required for a film. The fields are the same fields used for the rest of the BPL literary catalog. For example, the metadata for the director of the film is presented in the title of the entry and there is no director field within the metadata of any film. This is an important omission, as the director is considered the single most important creative author; there is no "author" to a film that corresponds to the author of a book, but the director and writer warrant equal importance within their medium. The metadata on BPL does not list an author but has an "added author" field which lists the important creatives who contributed to the film. It's important that these names are included in the catalog record, but lumped together in this way is not an accurate representation of how these people contributed to the film. On IMDB, each creative has its own appropriate field, including Director, Writer, Producer, Cast, etc.

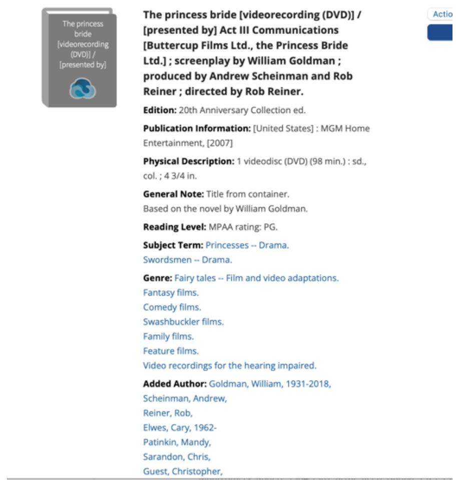

Fig. 1 Screengrab of BPL's metadata for the film The Princess Bride.

IMDB has appropriate fields for each important creative contributor to a film. Fig. 2 shows a sample metadata record. In addition to having appropriate fields, the metadata available on IMDB for each film is seemingly endless. Additional details for films include, but are not limited to: release date, production company, distributor, rating, technical specs, budget, production still photos, trivia, synopsis, tagline, parent guide, movie quotes, language, country of origin, box office numbers, and more.

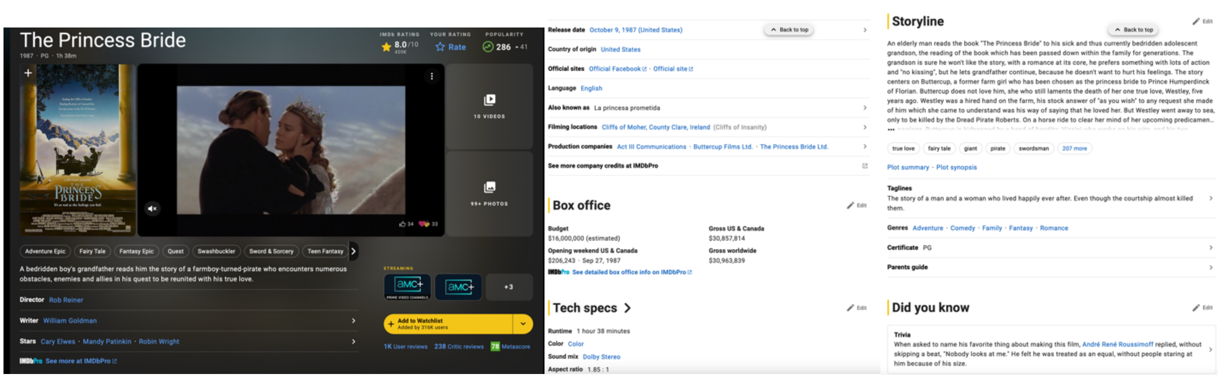

Fig. 2 Screengrabs of IMDB's metadata for the film The Princess Bride

Another important difference between these sites metadata records is the user generated information. On IMDB, the site allows reviews and user tags. The BPL catalog has a limited number of subject terms and genres, but IMDB has more expansive vocabulary to encapsulate the films aboutness. While the controlled vocabulary in BPL may be sufficient, the vocabulary used in IMDB is more complete. Many of the genres are similar in both BPL and IMDB, but IMDBs are more expansive and inclusive. A downside to the IMDB tags is its usability. Many films have over 300 user tags. Out of context, some of them may lack any meaning or pertinence.

A significant difference in the metadata, and perhaps the most important, is the format and physical description. BPL has a physical item to describe a call number, indicating that this film can be checked out by a patron, whereas IMDB does not. The metadata on BPL serves the purpose of pointing patrons to a physical item they can check out from the library. A finer point is made on this topic in looking at Exhibit A comparisons and noting that there are multiple copies of The Princess Bride at BPL that are different editions or releases of the same film. IMDB would not have any need to differentiate between different physical releases; BPL does have similar but slightly different metadata for their different physical copies of the same film, see Fig. 3 for comparison. IMDB metadata has much more data about a film but does not direct to a physical item. In FRBR terms, IMDB carries metadata for manifestations. However, a new feature on IMDB will link you to steaming services where users can rent or purchase the film digitally, marking a significant improvement in access point for IMDB. Within BPL, important format details are described within the physical description field, such as sound, color and aspect ratio. This is not an incorrect place to place this metadata, but a more correct location is in the Format field, which is where they are located on IMDB. A finer point is made on this topic in looking at Exhibit A comparisons and noting that there are multiple copies of The Princess Bride at BPL that are different editions or releases of the same film. IMDB would not have any need to differentiate between different physical releases, BPL does have similar but slightly different metadata for their different physical copies of the same film. While the BPL metadata is lacking, it is only needed insofar as it’s use, which is to find films that the library has available to check out. Further research would need to be conducted to determine if BPL patrons are finding and checking out films available in the library, and if the available metadata is sufficient for that use.

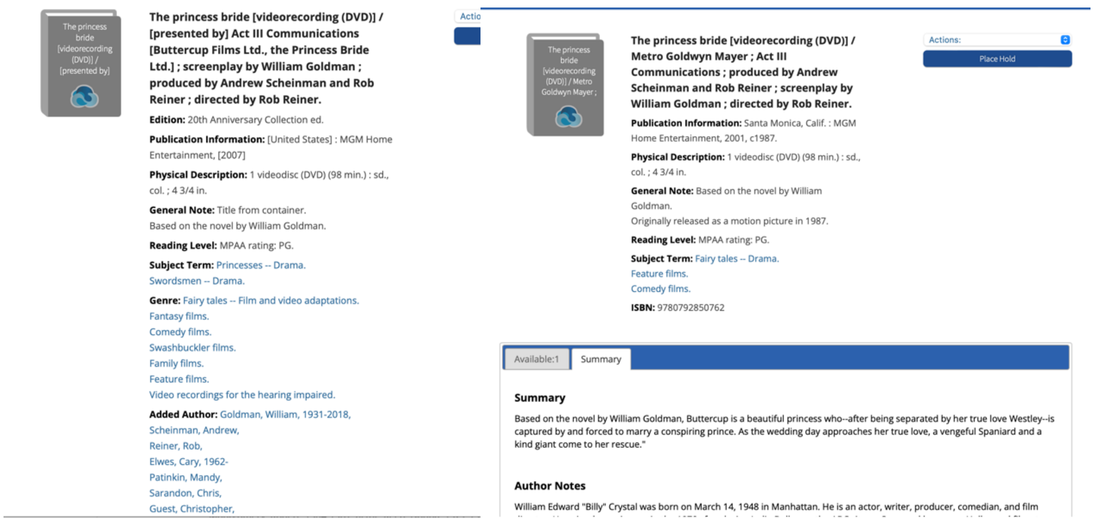

Fig.3 Screengrab of two different item records for the movie The Princess Bride.

It is important to note that even within film metadata, there is not a universal standard yet. In addition to this lack of standard is that no "single point of reference for metadata is the absence of a common vocabulary across the industry - very often there are different terms being used for the same data elements (Dexiades & Heath, p. 206). Even when a schema attempts to have their own standard, their terms may have different meanings to different users or groups. Soares and Viana note that "Although quite a lot of effort has been put on the standardization of a multimedia description schema, different solutions coming from different organizations and having different levels of details were published." (2014, p. 7028). One such organization, the International Federation of Film Archives, has created a standard called CWS (Cinematographic Works Standards) (Domínguez-Delgado, & Hernández, 2017, p. 255). These standards are starting to be used in archives and specialty libraries, but not universal or mainstream in places where moving images are not the main items being cataloged. This makes it hard to fault BPL for not using a better suited schema for their film metadata, especially when you consider that “metadata management in film archives has now become a field of its own, requiring specialized staff and professionals of various fields” (Ross and Eckes, 2017, p. 44). A public library may not have the resources or expertise to properly keep up with film metadata cataloging. However, the lack of needed access points for films within catalog systems for users cannot be understated. Having a cataloging system that can be flexible to these types of needs is imperative for proper cataloging of films if they are to be included in a general setting like this one.

Even within the eight films compared in this study, the BPL metadata is not consistent within its own schema. Rating, formats, titles and other fields are not consistent across these eight films metadata. What is clear is that BPL is not using any standard of film metadata, they are shoe-horning film metadata into their standard metadata schema used for the library as a whole, and one can assume mostly used for literary material. This is a deficiency within the libraries catalog and one that may leave patrons looking for films elsewhere.

What is most interesting about the metadata found on IMDB is its use beyond the database itself. It is not only used for a common movie consumer or movie industry professionals. As technology and streaming services have taken over the way people consume movies, their systems also has the ability to make user recommendations based on the metadata within its system. These systems are called Recommender Systems (RS) and are commonly used by streaming services who have the available resources to create algorithms with user and content metadata (Soares and Viana, n.d., p. 1).

This is an exciting development within metadata and user recommendation schemas. The companies utilize user rating with the metadata, but some have found these recommendations unsuccessful, Soares and Vianna have pointed out that “users might have rated equally the same movie but due to different reasons: either because of its genre, the crew or the director “ (n.d., p.2). They embarked on a study that compared the metadata used by Netflix with that of IMDB using two different algorithms in which they concluded “user ratings are correlated with metadata, bringing to surface the real reasons that drove user rating behavior” (Soares and Viana. n.d., p. 2). The study showed that the director of a film was the most important and common metadata point that corresponded to successful user recommendations (Soares and Viana, 2014, p. 7027). This highlights even more pointedly how BPLs lack of accurate fields, including no director field, is a significant deficit in their records.

When film metadata is created correctly, it has shown to be more useful than its analog counterparts. In a study done Maryland, a library chose to compare it’s microfilm records with its e-version records before weeding out it’s duplicate microfilm, and “it was discovered that the microfilm and e-version bibliographic records contained metadata which had not been utilized by OCLC to improve its link resolution and discovery services for digitized versions of the microfilm resources” (Guay, 2017, p.18). This is further evidence of the importance of accurately cataloging film metadata, and that it’s richness can improve systems in many ways beyond just data for the sake of data.

Film metadata is specialized enough to need a universal standard, but getting every local library resources to get up to par with such standard may take some time. Regardless, databases like IMDB do enable patrons interested in film metadata to find and access any information they might be seeking that they may be unable to locate through their local library. The world of film metadata is continuing to expand beyond just fun facts and access points, and recommender systems, and the rich metadata it utilizes, may continue to become more and more sophisticated and important as these universal standards hopefully become a reality.

Exhibit A

MOVIE METADATA

FILM	METADATA ORIGIN	CATALOG TITLE	DIRECTOR	ADDED AUTHOR	PUBLICATION INFORMATION	WRITER	ADDED CORPORATE AUTHOR	RELEASE	EDITION	SERIES	SUBJECT TERMS	GENRE	CALL NUMBER	FORMAT	PHYSICAL DESCRIPTION	RATING	NOTES
Cowboy Bebop	BPL	Cowboy bebop, the movie.	N/A	N/A	[United States] : Columbia TriStar Home Entertainment, 2003.	N/A	N/A	N/A	N/A	N/A	N/A	N/A	DVD COWBO	Visual Materials	1 videodisc (DVD) (114 min.): sd, col ; 4 3/4 in.	MPAA rating: R	N/A
Cowboy Bebop	IMDB	Cowboy Bebop: The Movie	Chin'ichiro; Tensai Okamura; Hiroyuki Okiura	N/A	Bandair Visual Company, Bones	Sin'ichiro Watanabe; Keiko	N/A	5/16/2003 (US)	N/A	N/A	terrorist, bounty hunter, neo noir,	Adult Animation, anime, action, animation, crime,	N/A	Running Time: 1 hr 55 min, Color, Sound	N/A	TV-14	Cast, Trivia, Box Office, Nominations
Raiders of the Lost Ark	BPL	Indiana Jones and the raiders of the lost Ar k[videorecording (DVD)] /	N/A	N/A	Hollywood, Calif; Paramount Home Entertainment c2003	N/A	Paramount Pictures Corporation, Lucasfilm, Ltd; Paramount Home	N/A	Special Ed.	Adventures of Indiana Jones	Jones, Indiana (Ficticious character --	Feature films, Action and adventure films;	DVD INDIA	Visual Materials	1 videodisc (DVD) (115 min.): sd, col ; 4 3/4 in.	MPAA rating: PG	Videodisc release of a 1981 motion picture. DVD-ROM
Raiders of the Lost Ark	IMDB	Raiders of the Lost Ark	Steven Speilberg	N/A	Paramount Pictures, Lucasfilm	Lawrence Kasdan, George Lucas,	N/A	June 12, 1981 (US)	N/A	Indiana Jones	nazi, ark of the covenant, egypt,	Adventure Epic, Desert Adventure, Quest, Action,	N/A	Runtime: 1 hr 55 min, Color, Dolby Atmos,	N/A	PG	Filming lcoations, Box office, cast,
Memento	BPL	Memento [videorecording (DVD)] / Newmarket presents in association with	N/A	Nolan, Christopher, 1970-; Todd, Suzanne; Todd,	Santa Monica, Calif. : Lions Gate Entertainment 2011	N/A	Newmarket Capital Group LP; Summit Entertainment; Team Todd (Firm); Lions Gate	N/A	N/A	N/A	Amnesia -- Drama; Short-term memory --	Feature Films. Video recordings for the hearing	DVD MEMEN	Video Disc	2 videodiscss (DVD)(77 min.): sound, color ; 4 3/4	MPAA Rating: Not ratedt	Based on characters created by Hanna-Barbera. 1080p high
Memento	IMDB	Memento	Christopher Nolan	N/A	Newmarket Capital Group, Team Todd, Memory productions	Christopher Nolan, Jonathan	N/A	May 25, 2001 US)	N/A	N/A	memory, short term memory loss, hidden	Mystery, Thriller	N/A	Runtime: 1 hr, 53 min; Color, Black and	N/A	Rated R for violence, language	Filming lcoations, Box office, cast, trivia, storyline,
The Grand Budapest Hotel	BPL	The Grand Budapest Hotel [videorecording (DVD)] / Fox Searchlight Pictures in	N/A	Anderson, Wes, Rudin, Scott; Rales, Steve; Dawson,	N/A	N/A	Fox Serchlight Pictures, Indian Paintbrush (Firm), Studio Babelsberg, American	N/A	N/A	N/A	Hotels -- Europe -- Drama; Hotel -	Dark comedy films, features filmes, Fuctionfilms,	DVD GRAND	Visual Materials			
The Grand Budapest Hotel	IMDB	The Grand Budapest Hotel	Wes Anderson	N/A	Fox Searchlight, Indian Paintbrush, Studio Babelsberg	Stefan Zweig, Wes Anderson, Hugo Guinness	N/A	March 28, 2014 (US)	N/A	N/A	hotel, lobby boy, painting, first person	Caper, Quirky Comedy, Adventure,	N/A	Runtime: 1 hr 39 min; Color, Black and	N/A	Rated R for alnguage, some sexual	Filming lcoations, Box office, cast, trivia, storyline,
The Princess Bride A	BPL	The princess bride [videorec ording (DVD)] / Metro Goldwyn Mayer ; Act III	N/A	Goldman, William 1931-2018; Scheinman, Andrew;	[United States] MGM Home Entertainment. [2007]	N/A	Act III Communications; Buttercup Films Ltd, Princess Bride Ltd, MGM Home	N/A	20th Anniversary Collection ed.	N/A	Princesses -- Drama; Swordsmen --	Fairy tales -- Film and video adaptations;	DVD PRINC	Video Disc	1 videodisc (DVD) (98 min.): sd, col ; 4 3/4 in.	MPAA rating: PG	Titles from container. Based on the novel by William
The Princess Bride B	BPL	The princess bride [videorec ording (DVD)] / Metro Goldwyn Mayer ; Act III	N/A	N/A	Santa Monica, Calif. : MGM Home Entertainment 2001, c	N/A	N/A	N/A	N/A	N/A	Fairy tales -- Drama; Feature films;	N/A	DVD PRINC	Visual Materials	1 videodisc (DVD) (98 min.): sd, col ; 4 3/4 in.	MPAA rating: PG	Based on the novel by William Goldman. Originally
The Princess Bride	IMDB	The Princess Bride	Rob Reiner	N/A	Act III Communications, The Brincess Bride Ltd, Buttercup Films, Ltd	William Goldman	N/A	October 9, 1987 (US)	N/A	N/A	true love, fairy tale, swordsmen,	Adventure Epic, Fairy Tale, Fantasy Epic, Quest,	N/A	Runtime: 1 hr 398 min; Color; Soundmix:	N/A	PG	Filming lcoations, Box office, cast, trivia, storyline,
The Rocketeer	BPL	The rocketeer [videorecordi ng (DVD)] / Walt Disney Pictures ; story by Danny	N/A	Johnston, Joe, 1950-; Campbell, Willian, 1959-; Arkin, Alan;	N/A	N/A	Walt Disney Pictures; Gordon Company' Silver Screen Partners IV; Walt Disney	N/A	20th Anniversary ed.Widescreen	Disney Blu-ray	Superheroes-- Drama; Good and evil --;	N/A	DVD ROCKE	Video Disc	1 videodisc (109 min.): sound, color ; 4 3/4 in.	MPAA rating: PG; some material may	Based on the graphic novel created by Dave
The Rocketeer	IMDB	The Rocketeer	Joe Johnston	N/A	Walt Disney Pictures, Touchstone Pitures, Silver Screen Parners IV	Dave Stevens, Danny Bilson, Paul De Meo	N/A	June 21, 1991 (UA)	N/A	N/A	1930s, deisel punk, steampunk, jet	Steampunk, Superhero, Action, Thriller, Sci-fi,	N/A	Runtime: 1 hr 48 min; Soundmix:	N/A	PG	Filming lcoations, Box office, cast, trivia, storyline,
To Catch A Thief	BPL	To catch a thief [videorecor ding (DVD)] / Paramount Pictures ; screenplay, John	N/A	N/A	Hollywood, Calif; Paramount Home Entertainment [2007]	N/A	Paramount Home Entertainment (Firm)	N/A	N/A	N/A	Jewel thieves -- Drama; Love -- Drama	Feature films; Detective and mystery films;	DVD TO	Visual Materials	1 videodisc (ca 106 min.): sd, cl ; 4 3/4 in.	Rating: Not rated	Originally released in 1955. Based on the novel by David
To Catch A Thief	IMDB	To Catch A Thief	Alfred Hitchcock	N/A	Paramount Pictures	John Michael Hayes, David Dodge, Alec	N/A	August 5, 1955 (US)	N/A	N/A	french riviera, jewel theif, caper film	Whodunnit, Mystery, Romance, Thriller	N/A	Runtime: 1 hr 46 min; Color, 1:85.1	N/A	PG for some action violence, mild	Filming lcoations, Box office, cast, trivia, storyline,

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